RESULTS: elliptic-code list decoding as a relation finder, R2 panel (slug r2ecldan)

Verdict (derived by the PROTOCOL section 7 rule, applied mechanically): UNRESOLVED. [pcc:unverified]
P1 holds; D1 costs more than both E1 and E2 per verified relation in every cell with k >= 3 at w = 0.1 and w = 1 (measured), but P4 (slope within 0.25/0.4 of k-1) fails in all four k, and P3's factor-10 clause fails in two cells; the ENUMERATION-IN-DISGUISE branch needs P4, so it is not reached, and BELOW ENUMERATION is not met (D1 is never cheaper). No claim about the Zhang-Liu procedure. This is a toy relation-phase measurement, not a leaderboard row.

Evidence classes: measured = read from panel.jsonl; derived = computed from measured rows by r2ecldan_analyze.py; every number is [pcc:unverified].

0. Execution vs ecld_jobs.txt (reported first)

- PROTOCOL.md sha256 = e8481f7c...ac4465 (matches the frozen value; verified before analysis). ecld_toy.rs sha256 de901e23... and ecld_jobs.txt sha256 d3bef30a... equal the values in binary.sha256/the freeze (measured).
- sha256sum -c SHA256SUMS: 132/132 OK, re-run after the coordinator's note that panel.jsonl was briefly absent (measured). My analysis read 696 rows; sort lines/*.jsonl and sort panel.jsonl have identical sha256 (measured).
- 64 job lines in ecld_jobs.txt (56 curve jobs + 8 per-target jobs for cell 5,256), 64 line_NN.jsonl (all non-empty), 64 .err files, all 0 bytes, progress.txt: lines 64 fail 0, every line exit 0 (measured).
- Rows: 696 = 232 instances x 3 arms. Expected from cells_full: 14 cells x 16 + 1 cell (5,256) x 8 = 232. Missing: none. Extra: none. Deviation from ecld_jobs.txt: none found (measured).
- Host: 2 vCPU AMD EPYC 9R45 pod (nproc 2), jobs run sequentially by the entrypoint; protocol said one thread per job, no co-running panel jobs (the entrypoint's concurrency not audited here). Wall time is a practicality note only.

P1 and controls (measured)

check	result
E1=E2=D1 as sets (sets_consistent_across_arms)	232/232 instances
verified == relations, all arms	232/232
planted control recovered by all 3 arms	58/58 planted instances
A/A aa_e1_identical (k<=4)	144/144 true; k=5,6 null by design (88/88)
D1 gae == 0	violations: 0
negative-control cells (k=2 all B, k=3 B=64): arms agree where enumeration finds none	arms agree on all instances (consistent 232/232). Random-target relations observed: (2, 64): 0 (exp 0.38); (2, 128): 0 (exp 1.53); (2, 256): 6 (exp 6.25); (3, 64): 11 (exp 7.96). Planted instances give the nonzero counts (not the null set); the control is about random targets.

P1: HOLDS. Random-word control: deferred in the protocol, not run (stated gap).

P2: D1 gae = 0 and native count within 25% of plan (measured vs plan)

k,B	n inst	D1 native / instance (mean)	plan	ratio	within 25%
2,64	16	6432	6368	1.0101	yes
2,128	16	2.515e+04	2.502e+04	1.0053	yes
2,256	16	9.946e+04	9.92e+04	1.0026	yes
3,64	16	2.468e+05	2.446e+05	1.0090	yes
3,128	16	1.862e+06	1.853e+06	1.0047	yes
3,256	16	1.444e+07	1.441e+07	1.0020	yes
4,64	16	6.573e+06	6.323e+06	1.0395	yes
4,128	16	9.209e+07	9.004e+07	1.0227	yes
4,256	16	1.364e+09	1.348e+09	1.0122	yes
5,64	16	1.36e+08	1.271e+08	1.0698	yes
5,128	16	3.512e+09	3.363e+09	1.0443	yes
5,256	8	9.781e+10	9.537e+10	1.0256	yes
6,32	16	4.823e+07	4.32e+07	1.1164	yes
6,48	16	4.603e+08	4.175e+08	1.1025	yes
6,64	16	2.273e+09	2.083e+09	1.0914	yes

P2: HOLDS (D1 gae = 0 on all 232 instances; max ratio 1.116 at k=6,B=32).

P3: priced cost, D1 against E1 and E2, k >= 3 (derived: cell totals over all instances)

Columns: D1/E1 and D1/E2 priced-cost ratios at w=0.1 and w=1; w* = E gae / D1 native (protocol definition) vs E1 and E2. Factor-10 clause applies at k >= 4 (all four ratios must be >= 10).

k,B	D1/E1 w.1	D1/E1 w1	D1/E2 w.1	D1/E2 w1	w* (E1)	w* (E2)	D1 > both	w*<0.1	>=10x (k>=4)
3,64	5.82	40.38	5.8	40.4	0.0163	0.01634	True	True	None
3,128	10.90	75.95	10.9	75.9	0.0087	0.00873	True	True	None
3,256	21.06	147.07	21.1	147.1	0.0045	0.00452	True	True	None
4,64	7.34	51.72	101.9	652.1	0.0130	0.00092	True	True	False
4,128	12.70	89.20	354.0	2265.9	0.0075	0.00026	True	True	True
4,256	23.38	163.93	1306.4	8360.7	0.0041	0.00007	True	True	True
5,64	9.87	69.73	148.2	1037.6	0.0097	0.00064	True	True	False
5,128	15.44	108.55	478.5	3349.4	0.0062	0.00020	True	True	True
5,256	26.45	185.54	1665.8	11660.4	0.0036	0.00006	True	True	True
6,32	10.39	74.71	286.0	1869.9	0.0092	0.00033	True	True	True
6,48	12.09	86.04	799.1	5187.5	0.0079	0.00012	True	True	True
6,64	13.62	96.52	1655.1	10705.6	0.0070	0.00006	True	True	True

P3: FAILS as written. The ordering clause and w* < 0.1 hold in all 12 cells. The factor-10 clause fails at (4,64) (D1/E1 = 7.34 at w=0.1) and (5,64) (9.87 at w=0.1); at w=1 all are >= 10. E1 is the binding comparator; against E2 the factor is >= 101.9 everywhere. No threshold changed.

P4: slope of D1 native count vs B (derived)

k	B range	slope (endpoints)	LS slope over all B	pre-specified k-1	tolerance	holds
3	64-256	2.935	2.935	2	0.25	NO
4	64-256	3.849	3.849	3	0.25	NO
5	64-256	4.745	4.745	4	0.25	NO
6	32-64	5.559	5.559	5	0.4	NO

P4: FAILS in all four k (slopes 2.94, 3.85, 4.75, 5.56 against 2, 3, 4, 5). Heuristic observation, not a re-binning: the protocol's own P2 cost formula C(B,k-1) x [(B-k+1)(2k-1)/k + ...] contains a factor (B-k+1), so it scales as B^k, and the measured slopes follow that (they sit below k only through lower-order terms). The prediction text 'B^(k-1)' is inconsistent with the protocol's own formula; the prediction is nonetheless scored as written. C3 (D1 grows as B^(k-1)) is therefore not supported; the observed growth is about B^k.

P5: relations per random target vs C(B,k)/N (measured vs derived)

Pooled over random targets (3 per curve) using each curve's own N. Evaluable rule: expected total >= 10; ratio within [0.5, 2]. Below 10 the protocol checks 'direction only' with no numeric rule, so those cells are listed, not scored.

k,B	observed verified (random)	expected	ratio	evaluable	holds
2,64	0	0.4	0.00	False	not evaluable (direction only)
2,128	0	1.5	0.00	False	not evaluable (direction only)
2,256	6	6.2	0.96	False	not evaluable (direction only)
3,64	11	8.0	1.38	False	not evaluable (direction only)
3,128	88	65.0	1.35	True	True
3,256	509	528.4	0.96	True	True
4,64	122	118.8	1.03	True	True
4,128	2044	2010.7	1.02	True	True
4,256	33657	33653.1	1.00	True	True
5,64	1513	1458.8	1.04	True	True
5,128	50737	50820.3	1.00	True	True
5,256	843199	843442.5	1.00	True	True
6,32	185	172.7	1.07	True	True
6,48	2347	2367.9	0.99	True	True
6,64	14386	14298.2	1.01	True	True

P5: HOLDS on the 11 evaluable cells (ratios 0.96-1.35, mostly 0.96-1.07); (2,64), (2,128), (2,256), (3,64) have expected < 10 and are not evaluable by the numeric rule (observed 0, 0, 6, 11 against 0.4, 1.5, 6.2, 8.0).

Decision rule (PROTOCOL section 7)

- P1 holds: yes. Cells with k >= 3 and expected relations >= 1 per target: 3,128, 3,256, 4,64, 4,128, 4,256, 5,64, 5,128, 5,256, 6,32, 6,48, 6,64. In each, D1 cost per verified relation > E2's at w = 0.1 and 1: True (also > E1's in all 12 k>=3 cells, measured).
- Slope of P4 holds: no. So ENUMERATION IN DISGUISE is not reached by the rule.
- BELOW ENUMERATION (D1 cheaper than both in any k>=3 cell at both w): no cell.
- Outcome: UNRESOLVED (pattern 'any other', here driven by P4, with the cost ordering favouring enumeration everywhere). No win for either side is declared by the rule. Derived summary only: nothing in these rows shows D1 cheaper than enumeration; the formal discriminator is not satisfied because the slope clause failed.

Cost / operation-count figure

SVG: results/R2/analysis/r2ecldan_fig1.svg. Left: priced cost per instance at w=0.1 (derived). Middle: D1 native count vs B with fitted slopes. Right: relation yield, observed vs expected.

Relation-phase distance to rho is not computed here (protocol: toy, no replay certificate/validator/matched measured rho; not a leaderboard row).

experiment.export/v1

Not assembled. tools/experiment_protocol.py (crypto-autoresearcher) from-rows/check-export need a machine-readable experiment.protocol/v1 JSON; PROTOCOL.md states that record is for the experiment director to produce and none exists in the study dir. from-ecbench/from-taskq do not fit this command-job output. Skipped, not forced.

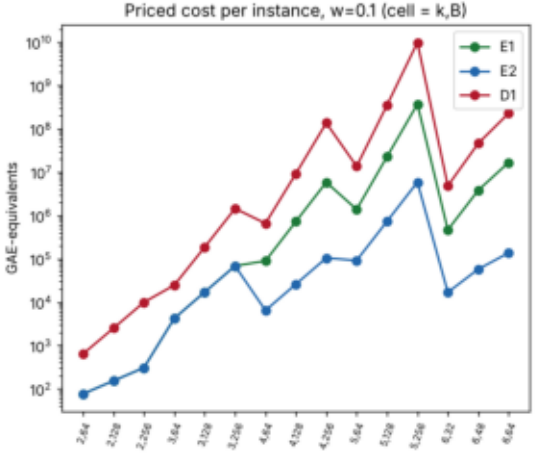
Provenance of this analysis

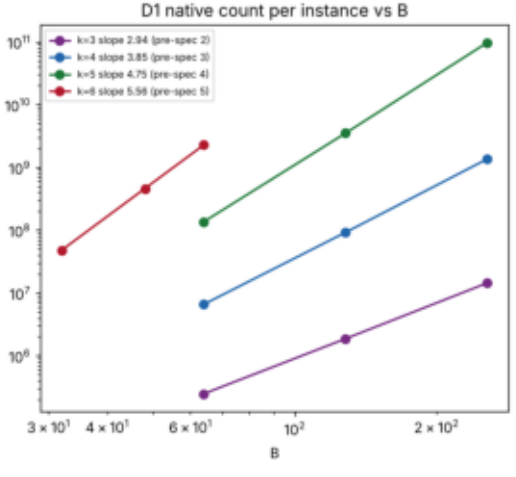
AGENTS.md forbids Python for research result generation, so these scripts are not committed; they live in the scratchpad (used only to tabulate/plot recorded values; a native re-derivation would be needed before any promotion). sha256:

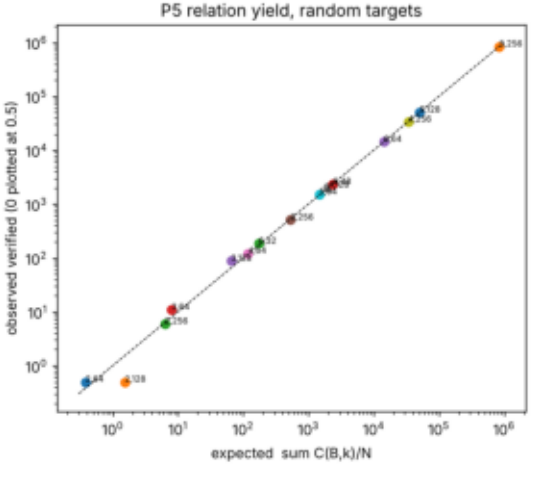
- r2ecldan_analyze.py 54f11359b2f8e7a5af40ea4631af3997c601544e8ec5ffc05213f208938b6589
- r2ecldan_fig.py 908bc2dce3111f91a05081166a9b52a26256cabba5f4ad9dee2315ce6b585ed8
- r2ecldan_md.py (this generator; hash not self-referential)
- results/R2/analysis/r2ecldan_summary.json b094e2e0dca7dafaa0347767fd6c5148fc84cb47f8a5b65e9a1aaf4c2bb5e8de
- Raw: results/R2/c24ac183-0e85-4a80-a46b-b51eb4db180d/panel.jsonl (SHA256SUMS verified).

Sources checked: research-kb and Conductor are not installed in this environment; none queried.

PDF: built with matplotlib's PDF backend (no typst, no LaTeX; pandoc present but no PDF engine).


Priced cost per instance, w=0.1 (cell = k,B)
GAE-equivalents
E1
E2
D1


D1 native count per instance vs B
k=3 slope 2.94 (pre-spec 2)
k=4 slope 3.85 (pre-spec 3)
k=5 slope 4.75 (pre-spec 4)
k=6 slope 5.56 (pre-spec 5)
B


P5 relation yield, random targets
observed verified (0 plotted at 0.5)
expected sum C(B,k)/N